

Jiahui Li

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Personal Profile

A self-motivated and passionate master graduate at UCL Computer Science, eager to pursue a Ph.D. in reinforcement learning. My experience spans work as a game designer for mobile title *Guns of Glory* at *FunPlus*, and a research experience training a RL agent in a Unity game to surface game design insights. I am currently interested in multi-agent RL, with a special focus on applications in games.

Education

University College London, Department of Computer Science London, UK
MSc in Computer Science Oct 2024 – Dec 2025
Key Modules: Applied Deep Learning, Artificial Intelligence and Neural Computing, Algorithmics

University of Cambridge Cambridge, UK
Mphil in Technology Policy Oct 2020 – Jul 2021
Capstone project on Algorithmic Governance with the *Paris Peace Forum*

University of Liverpool Liverpool, UK
BSc in Mathematics, 1st Class 2016 – 2020
2+2 programme with Xi'an Jiaotong-Liverpool University, Suzhou, China

Research Experiences

Deep RL for Weapon Design insights in a 2D arena-combat Game 2025
MSc Dissertation, University College London — supervised by Prof. Robert Graham
[Video], [Code]

- Designed and developed a battle survival game in Unity C# following a studio-standard process such as a game design document, stats tuning, and art asset specs. Implemented and passed unit tests against a MoSCoW requirement list for this game.
- Built an environment wrapper and training pipeline on the game using Unity ML-Agents, and implemented PPO-clip from scratch in PyTorch. Trained an agent to move and deploy weapons tactically for maximum survival and provided an evaluation script for logging and video capture.
- Resolved learning plateaus through reward design and hyperparameter tuning: a wall-proximity penalty to prevent corner-camping, a tuned discount factor for delayed payoffs, and an exploration schedule that surfaced kiting behaviour. Accelerated learning 10× and revealed weapon synergy insights.

Projects

PPO implementations on MuJoCo locomotion control [Code]
Reference: Schulman et al., *Proximal Policy Optimization Algorithms*, 2017

- Rebuilt the ablation study from the openai PPO paper using vanilla PG, PPO-clip, and PPO with KL penalty, built on a shared GAE actor-critic backbone referencing *CleanRL*. Improved the baseline performance by adding a bootstrap term counting for fake termination due to time limit truncation.
- Ran a controlled comparison on MuJoCo HalfCheetah-v4 and Ant-v4. Regained paper's result that trust-region methods outperform vanilla PG in terms of both speed and performance, while PPO-clip learning faster early on and adaptive-KL converging higher later in training.

Multi-Agent PPO on Cooperative MPE Benchmarks

[Code]

Reference: Yu et al., *Effectiveness of PPO in Cooperative Multi-Agent Games*, NeurIPS 2022

- Reproduced Yu's results on MPE cooperative benchmarks using CTDE architecture and recurrent multi-agent (rMAPPO) algorithm from the paper.
- Implemented the full training pipeline including a vectorized environment wrapper, a rollout buffer supporting BPTT, a GRU-based recurrent network backbone, and a PopArt value normalization head.
- Run experiments on scenarios including simple-reference -40 to -13 over 12M steps and simple spread -220 to -115 over 40M steps with video illustration, matching paper's results and its finding: PPO methods can achieve competitiveness after fine-tuning under cooperative MARL setting.

Industry Experience

Game designer(graduate program)

2022 – 2024

FunPlus Kings Group, Beijing

- Designed game monetisation systems for live mobile title *Guns of Glory*(50M+ downloads), including mechanics design, numerics configuration, and UI optimisation to encourage player spending.
- Wrote SQL and built visualisation charts using proprietary BI tools to analyse user spending metrics, such as ARPDAU, CTR, LTV etc. Identified trends by users' spending tiers, and delivered written reports for system improvement.
- Built an automated configuration tool replacing manual data entry for game system setup, cutting configuration time by 90% and eliminating manual-entry errors.

Honors and Awards

- Frank Stanton Carey Prize in Mathematics at University of Liverpool, departmental award for outstanding academic performance (£1500)
- FunPlus Game jam, shipped a complete game with a 5-person team in 48 hours, contributing C# gameplay code and Stable Diffusion art assets (ranked 4/12)

Relevant Research Skills

Programming Language: Python, C#, SQL, C++

Frameworks and Tools: NumPy, Pandas, PyTorch, Gymnasium, Git, Docker

Game Developmen: Unity, Unreal

Language: IELTS Listening 9.0, Reading 8.5, Writing 7.0, Speaking 7.0